

Jiashuo Chang

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Education

Nanyang Technological University

Master of Computing in Applied Artificial Intelligence (MCAAI),
College of Computing and Data Science (CCDS)

08/2026–Present

Central South University

Bachelor's Degree in Computer Science and Technology

09/2022–07/2026

- Final weighted average score: **90.95/100**.

University of Dundee

Dundee International Institute of Central South University

09/2022–07/2026

- Awarded **First Class Honours**.

Research Projects

MMP-2K | Macro Photography IQA Benchmark

IEEE ICIP 2025 [Paper](#) [GitHub](#)

02/2024–01/2025

- Led **dataset design, image selection, subjective studies, and distortion-oriented annotations** for macro-photography quality issues including depth of field, out-of-focus blur, and subject detail.
- Designed **subjective-scoring, quality-control, and label-verification workflows**; developed a **Vue + Flask** platform for annotation, sample retrieval, study management, and result viewing, with resources released on GitHub and Hugging Face.
- Delivered a **reproducible macro-photography IQA benchmark** that exposes limitations of generic IQA metrics in this setting; presented the work as a Poster Presenter at IEEE ICIP 2025.

FLEX-MMP | Explainable Multi-task Macro Photography IQA

Ongoing Research Project

04/2026–Present

- Develop **FLEX-MMP**, an explainable multi-task NR-IQA framework for authentic macro-photography distortions, using **multi-scale ResNet quality features** as a shared representation so global quality regression and local diagnosis jointly update the core feature layers.
- Design branches for **distortion presence, 3×3 spatial location, severity, and a weak degradation map**; use presence-masked location/severity losses, Focal Loss, and uncertainty-weighted map supervision for multi-label imbalance and weak spatial labels.
- Achieve **PLCC 0.880 and SRCC 0.800** in five-fold MMP-2K validation; use comparative and ablation studies to assess the contribution of multi-task supervision to quality prediction, diagnosis, and generalization.

When Composition Doesn't Add Up | Human Detection of AI Image Defects

IEEE MMSP 2026 [Paper](#) [GitHub](#)

08/2025–09/2026

- Constructed the **CO-AID study dataset**, organizing generation prompts and AI-generated images; designed **global defect categories and localized annotations** with normalized coordinates for traceable image-participant-defect records.
- Led **modeling and validation** for a human study of compositional defects in AI-generated images, combining participant-reliability assessment and outlier handling to analyze relationships between defects and human perception.
- Produced reproducible data and analysis workflows for systematic comparison and follow-on studies of defects in AI-generated images.

- Conducted **surveys and interviews** to identify intervention needs, user expectations, and functional boundaries for AI psychological assistants.
- Facilitated low-fidelity **participatory design** with psychology and computer-science students, then collected feedback through a high-fidelity web prototype to translate findings into evidence-based system requirements.

Internship Experience

Western Securities Co., Ltd.

IT Intern, Information Technology Department 07/2025–08/2025

- Analyzed securities workflows and mapped data sources, requirements, and application scenarios for **AI-enabled operational support**, providing business-side input for solution design.
- Built **Dify-based AI agent prototypes** for multi-source data integration and multi-agent fault analysis, validating the feasibility of agent-based business assistance.
- Contributed to **system modeling, data preparation, and technical-solution design** for internal AI competition projects, translating business requirements into demonstrable prototypes and implementation paths.

EAST Energy Storage Technology Co., Ltd.

Backend Intern, Photovoltaic and Energy Storage Products Department 07/2024–09/2024

- Implemented **voice and face-recognition login** using Baidu AI Cloud APIs and a Python/Flask backend, integrating third-party AI services with the product authentication flow.
- Developed **API integration and data-processing flows** for an energy-storage product platform, mapping service dependencies and improving feature completeness and maintainability.
- Diagnosed integration and data-flow issues during delivery, turning external AI capabilities into maintainable product modules.

Awards and Skills

- Third-Class Scholarship, Central South University 2022–2023, 2023–2024, 2024–2025
- Outstanding Graduate, Central South University 06/2026
- Excellent Student Leader, Central South University 2022–2023
- Third Prize, China Collegiate Service Outsourcing Innovation and Entrepreneurship Competition 06/2024
- **Programming:** Python, Java, C, C++, JavaScript, HTML/CSS, Bash
- **Tools:** Flask, Vue, ElementUI, Dify, Git, Codex
- **Languages:** Mandarin Chinese native; English IELTS 7.0 (L 8.5 / R 7.5 / W 6.0 / S 6.5), CET-4 589, CET-6 515